

COMP6237 Data Mining

Lecture 12: Outlier Detection

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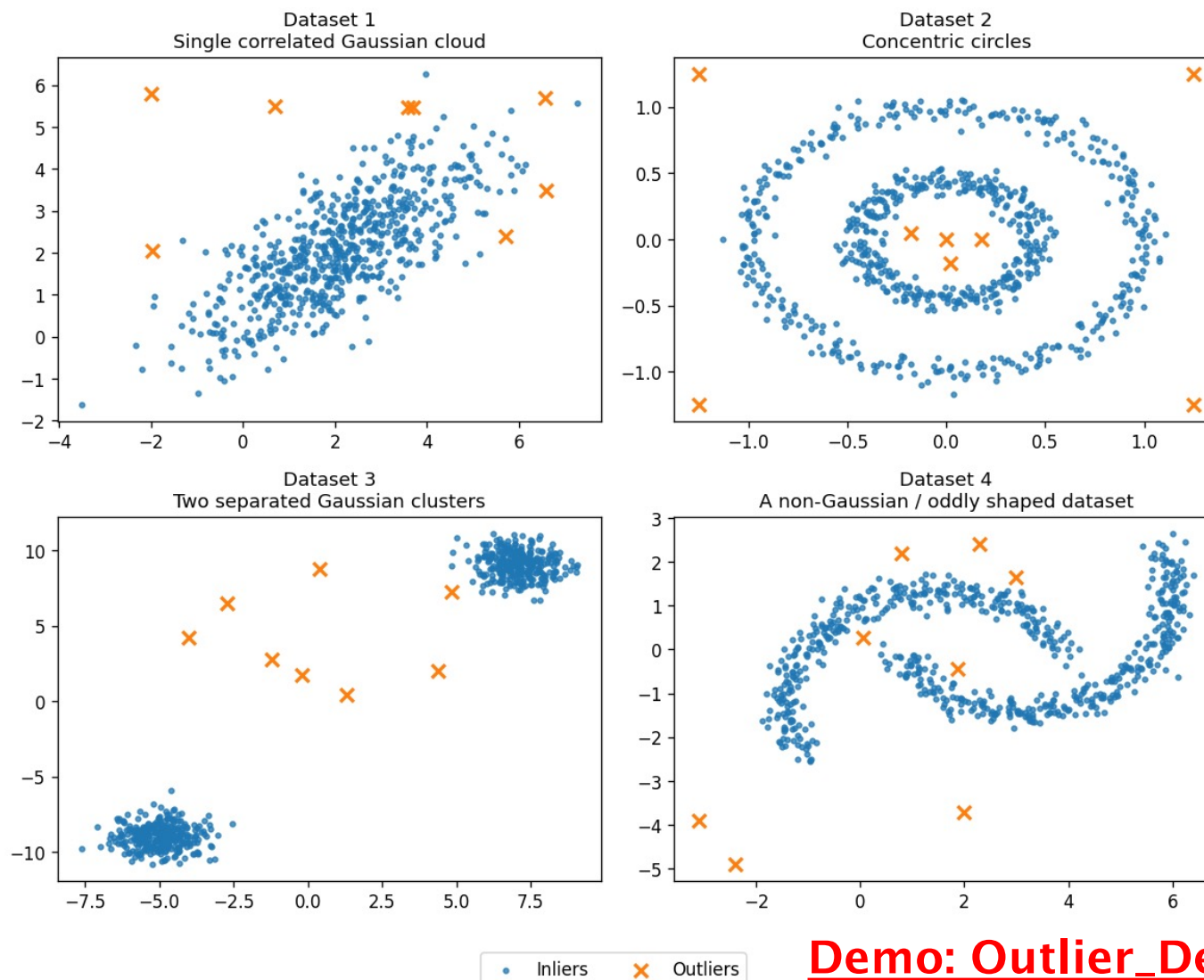
Lecture slides available here:

<http://comp6237.ecs.soton.ac.uk/zh.html>

(Thanks to Prof. Jonathon Hare and Dr. Jo Grundy for providing the lecture materials used to develop the slides.)

Outlier Detection – Problem

Outlier detection aims to identify observations that deviate significantly from the normal pattern of the data. An **outlier** is a data point that is inconsistent with the pattern followed by most observations.



Demo: Outlier_Detection_demo

Outlier Detection – Problem

Extreme values are not outliers!

Extreme Value Statistics

A way to characterise extreme values using a rule similar to the central limit theorem.

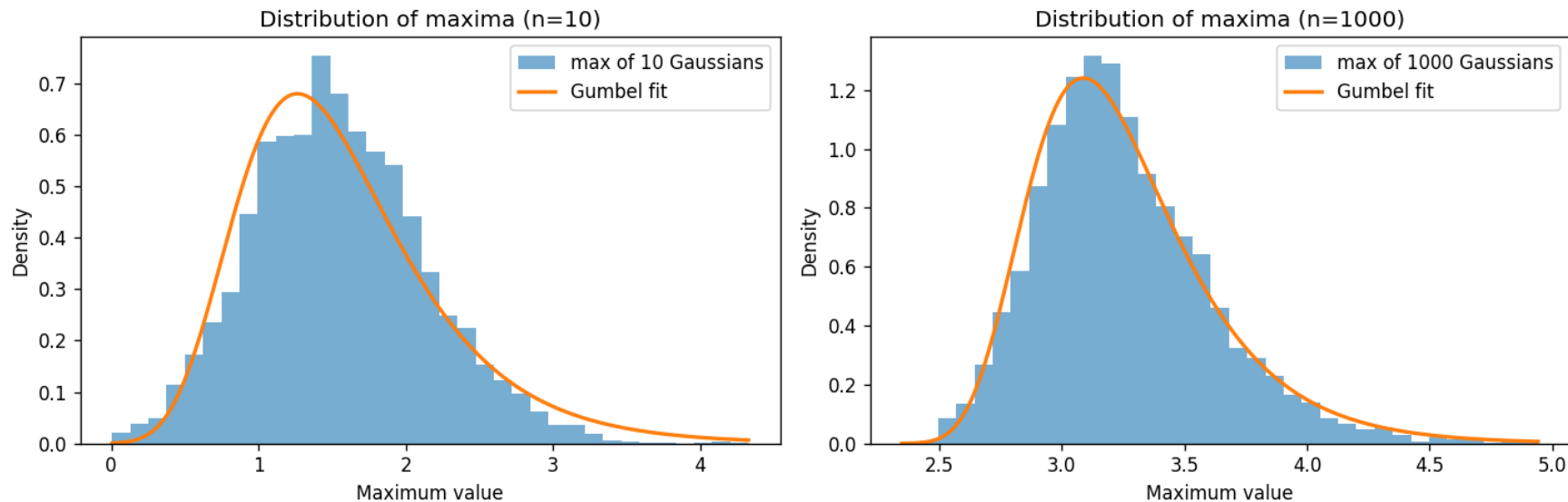
Also known as the Fisher-Tippett theorem

$$f(x) \approx \frac{1}{\beta} e^{\frac{x-\mu}{\beta}} - e^{\frac{x-\mu}{\beta}}$$

- The limiting distribution of these extremes must be one of three types of generalized extreme value distributions - Gumbel, Fréchet or Weibull
- It makes a distinction between **real anomalies** versus values that are simply **extremes** of the primary data distribution.

<http://www.nematrian.com/Pages/ExtremeValueTheoryCombined.pdf>

Outlier Detection – Problem



The Gumbel distribution is used here to give a probability that a value is an maximal value from a normal distribution. With more samples, the distribution is more clearly defined.

Demo: Outlier_Detection_demo

See e.g. S.J.Roberts IEE Proceedings 2000, 147,6,363-367

<https://ieeexplore.ieee.org/document/889967>

Outlier Detection – Approach

We can model data using a Gaussian distribution:
Univariate:

$$p(x) = \frac{1}{2\sqrt{2\pi}} \exp \frac{-\frac{1}{2}(x - \mu)^2}{\sigma^2}$$

Estimate mean:

- ▶ $\mu = \frac{1}{N} \sum_{i=1}^N x_i$

Estimate standard deviation:

- ▶ $\sigma = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)$

Outlier Detection – Approach

A Data mining approach:

- ▶ Model the data
- ▶ What does not fit is outlier

Can use many different models

Need:

- ▶ a measure of fit

Outlier Detection – Gaussian Distribution

We can model the data using a multivariate Gaussian distribution:

$$p(\mathbf{x}) = \frac{1}{2\pi^{\frac{p}{2}}\sqrt{|\mathbf{C}|}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \mathbf{m})^T \mathbf{C}^{-1}(\mathbf{x} - \mathbf{m})\right\}$$

Covariance and mean can be estimated from the data.. how?

$$\text{mean} = \mathbf{m} = \frac{1}{N} \sum_i^N \mathbf{x}_i$$

covariance is proportional to the inner product of the mean centred data

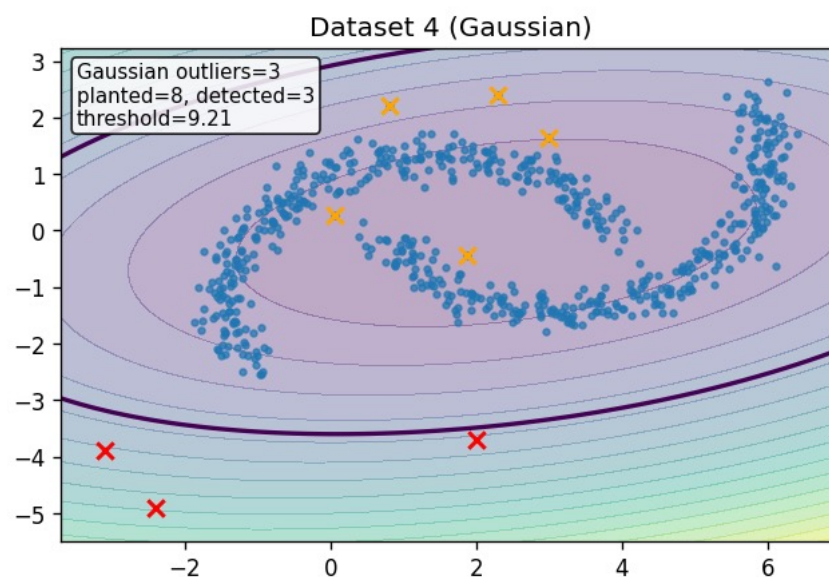
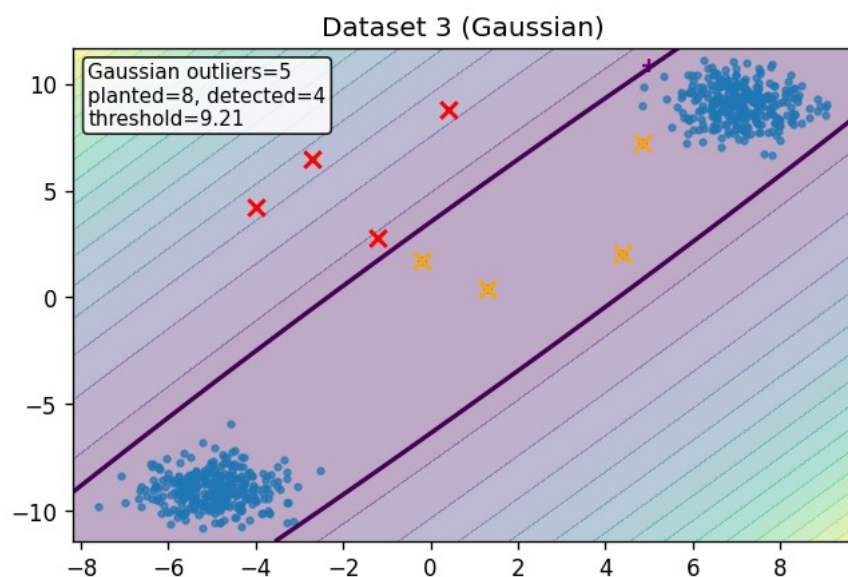
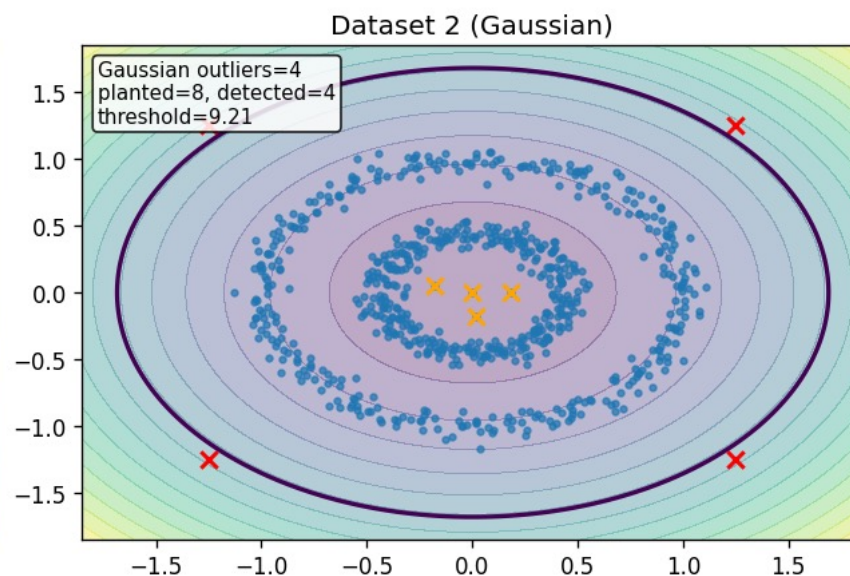
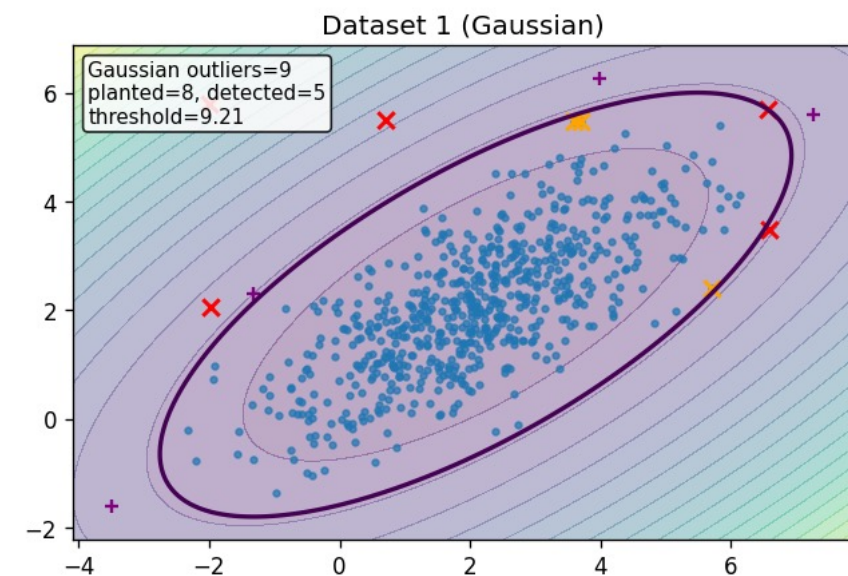
or

$$\mathbf{C} = \frac{1}{N} \sum_i^N (\mathbf{x}_i - \mathbf{m})(\mathbf{x}_i - \mathbf{m})^T$$

After fitting the multivariate Gaussian, measure how well a point fits the data cloud using Mahalanobis distance or Gaussian probability.

- Extreme value: unusual, but still plausible under the fitted distribution
- Outlier: too unlikely under the fitted distribution, beyond a chosen threshold

Outlier Detection – Gaussian Distribution



Demo: Outlier_Detection_demo

• regular points
 X planted outlier detected
 + extra Gaussian outlier
 X planted outlier missed

Outlier Detection – Gaussian Mixture Model

Try using more than one Gaussian: **Gaussian Mixture Model**

$$\sum_k^K \pi_k p(x|\mu, C)$$

Estimate weighting π , mean μ and covariance C ?

If we knew the weights, mean and covariance, we could calculate the probability

if we knew the probabilities, we could calculate the weights, mean and covariance

Expectation maximisation: generalisation of K Means

Outlier Detection – Gaussian Mixture Model

$$L(\theta|X_1, \dots, X_n) = \prod_{i=1}^n \sum_{k=1}^K \pi_k N(x_i; \mu_k, \sigma_k^2)$$

Algorithm 1: GMM

Data: X ($n \times p$ data), k Gaussians to use

Initialise π_k , μ_k and C_k ;

$$\ell(\theta) = \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k N(x_i; \mu_k, \sigma_k^2) \right)$$

while not converged **do** $\leftarrow$ Evaluate the log-likelihood with the new parameter

for $x_i \in X$ **do** $\leftarrow$ E-step: Evaluate the posterior probabilities

for $j \in 1, \dots, k$ **do**

 responsibilities $r_{i,j} = p(x_i | \mu_j, C_j)$;

end

end

for $j \in 1, \dots, k$ **do** $\leftarrow$ M-step: Estimate new parameters

$N_j = \sum_{i=0}^n r_{i,j}$;

$\pi_j = \frac{N_j}{N}$;

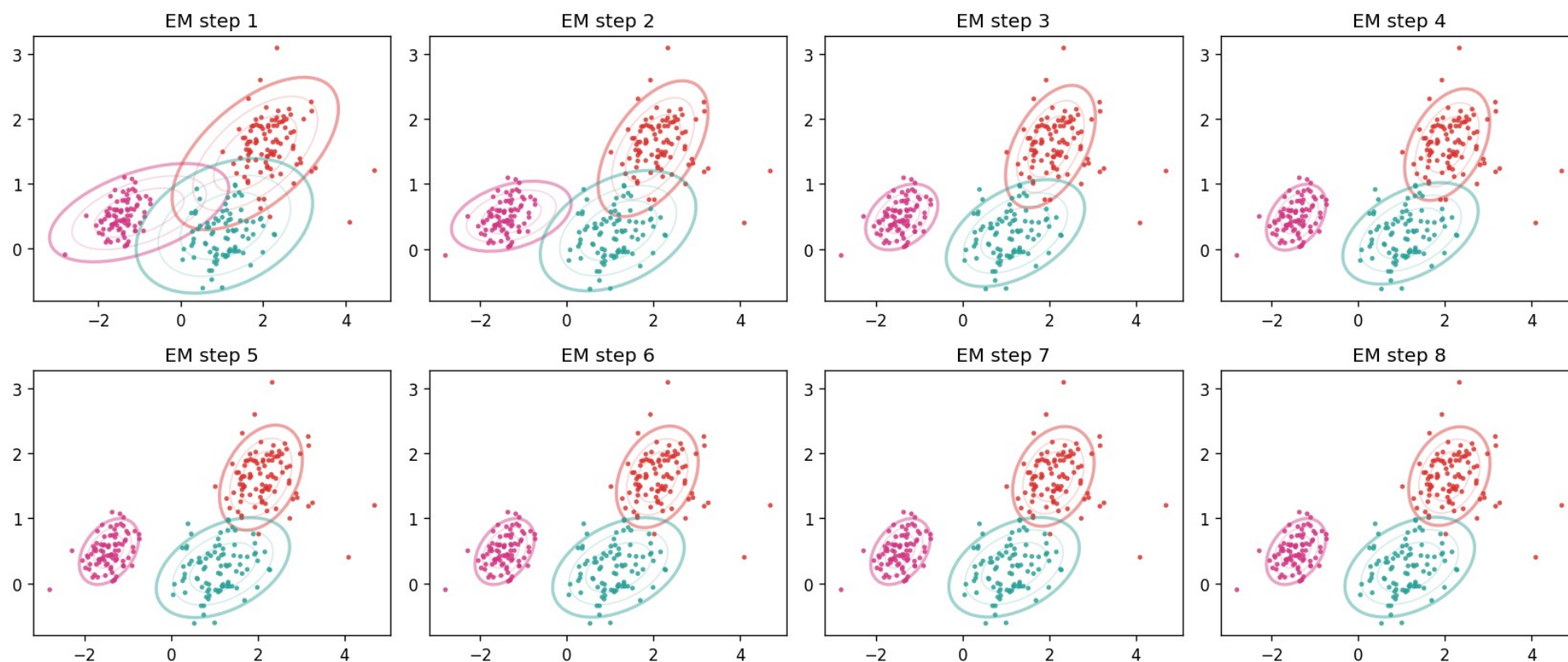
$\mu_j = \frac{1}{N_j} \sum_{i=0}^n r_{i,j} x_i$;

$C_j = \frac{1}{N_j} \sum_{i=0}^n r_{i,j} (x_i - \mu_j)(x_i - \mu_j)^T$;

end

end

Outlier Detection – Gaussian Mixture Model



Demo: Outlier_Detection_demo

Outlier Detection – Gaussian Mixture Model

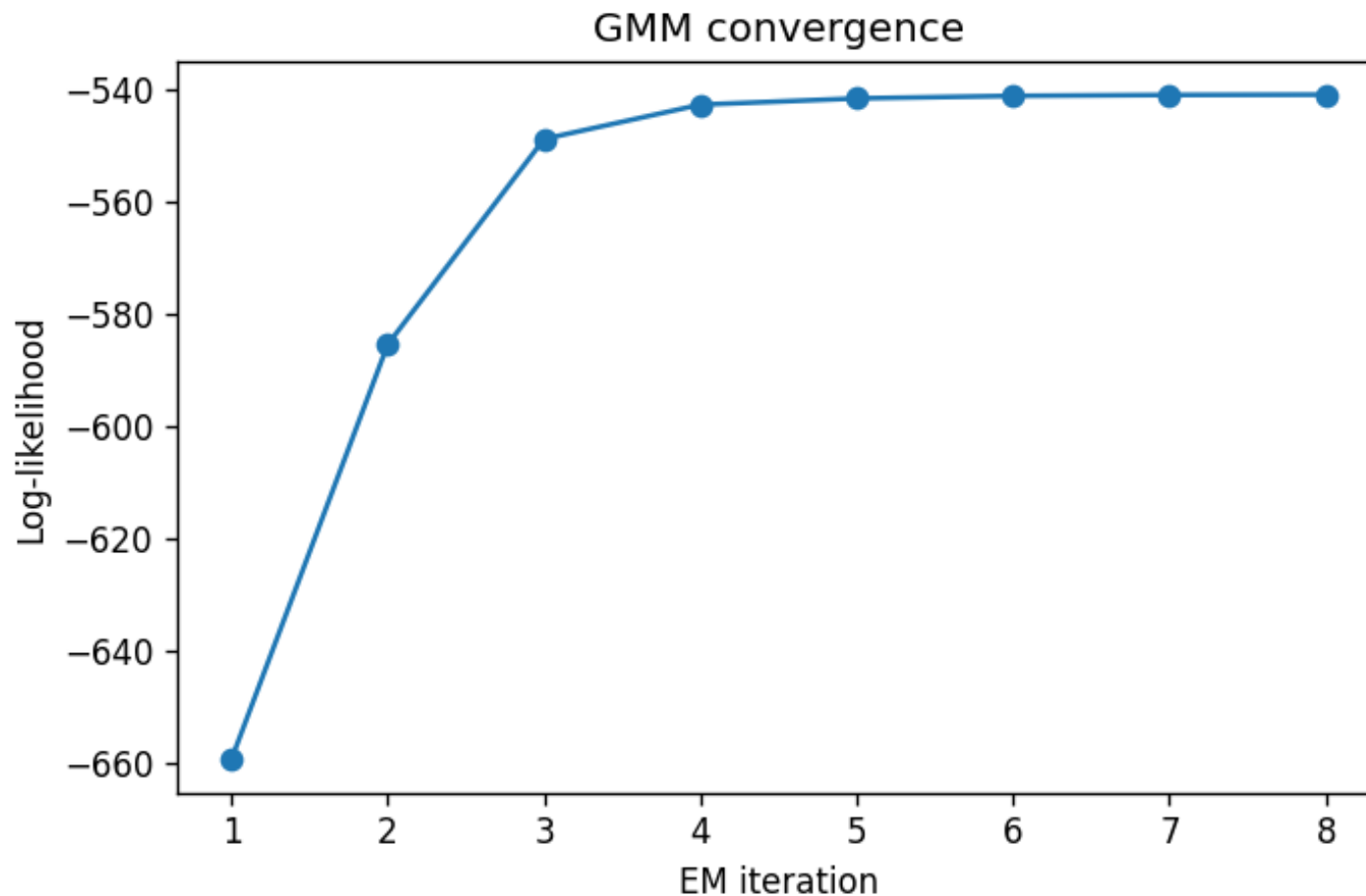
Initialisation:

- ▶ randomly - can cause issues
- ▶ use K Means - works quite well

Convergence:

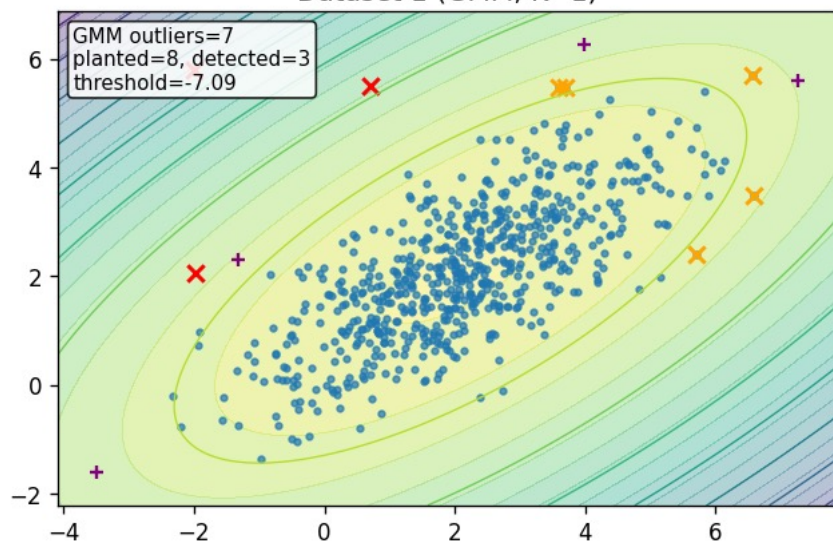
- ▶ Can check for an increase in the total probability
- ▶ $\sum_{i=0}^k \sum_{j=1}^n r_{i,j}$ $\ell(\theta) = \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k N(x_i; \mu_k, \sigma_k^2) \right)$
- ▶ best to use logs

Outlier Detection – Gaussian Mixture Model

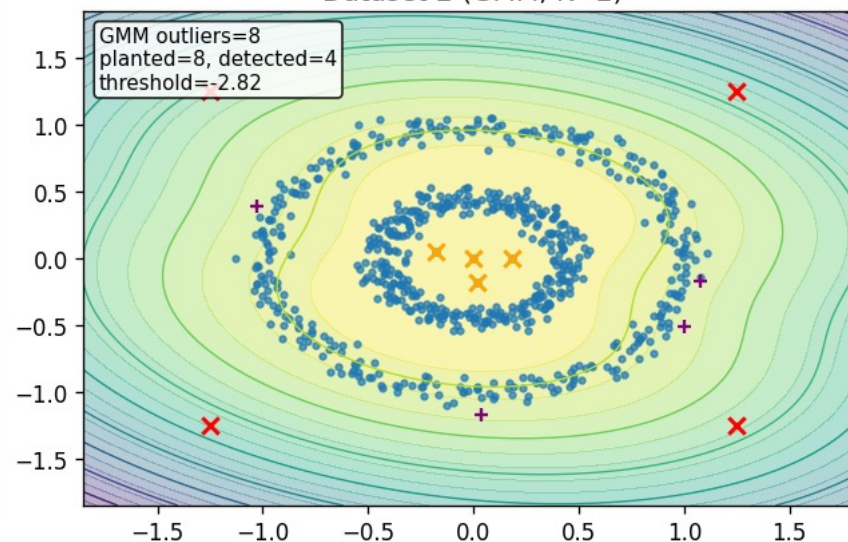


Outlier Detection – Gaussian Mixture Model

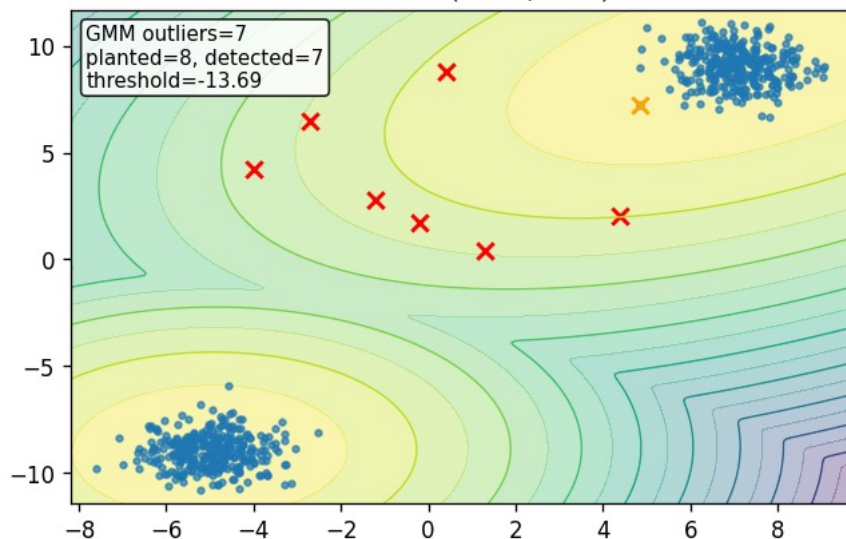
Dataset 1 (GMM, K=1)



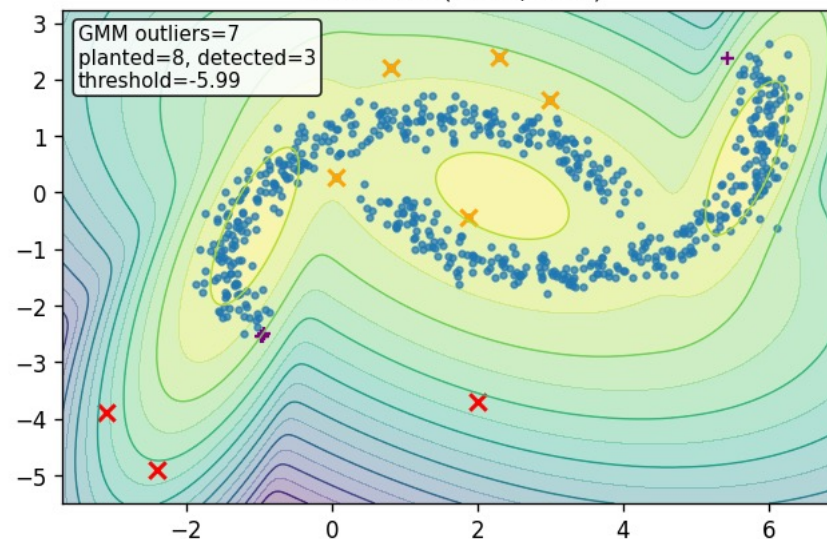
Dataset 2 (GMM, K=2)



Dataset 3 (GMM, K=2)



Dataset 4 (GMM, K=3)



Demo: Outlier_Detection_demo

• regular points X planted outlier detected + extra GMM outlier X planted outlier missed

Outlier Detection – DBSCAN

DBSCAN - good for outlier detection as well as clustering

Recap: Density Based Spatial Clustering and Noise

Needs:

- ▶ maximum radius $N_{\epsilon}(\mathbf{x}) = B_d(\mathbf{x}, \epsilon) = \{\mathbf{y} \mid \delta(\mathbf{x}, \mathbf{y}) \leq \epsilon\}$
- ▶ minimum number $|N_{\epsilon}(\mathbf{x})| \geq \text{minpts}$

Max radius is the limit on which to look for neighbours

Min number is the lower limit on what can be in a cluster

Outlier Detection – DBSCAN

Algorithm 2: DBSCAN

Data: X , ϵ , min_pts

initialise *labels* list as zeros, *count* list, *core* list;
Find neighbours for each point, Find core points;

$class = 1$;

for each core point p **do**

 add neighbours(p) to queue;

while queue not empty **do**

 neighbours = next(queue);

for q in neighbours **do**

 set label(q) = $class$;

if label(q) is 'core' **then**

 add neighbours(q) to queue

end

end

end

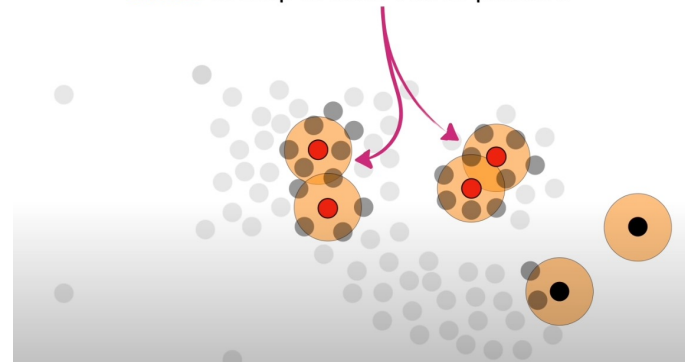
$class = class + 1$

end

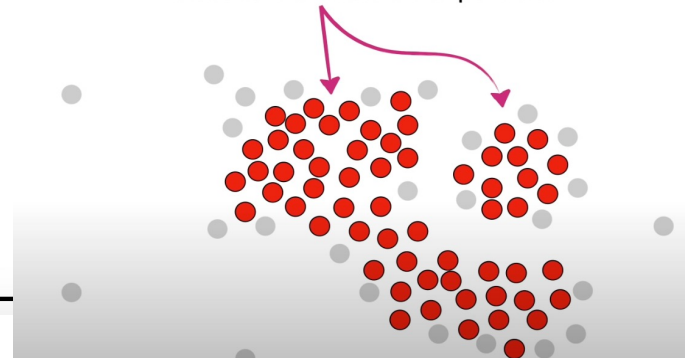
return labels;

$$|N_{\epsilon}(\mathbf{x})| \geq minpts$$

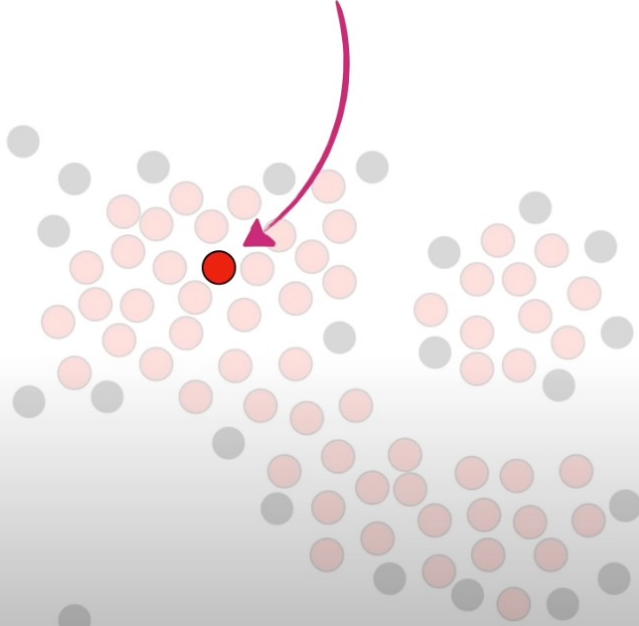
Anyway, these 4 points are some of the **Core Points**, because their **orange circles** overlap at least 4 other points...



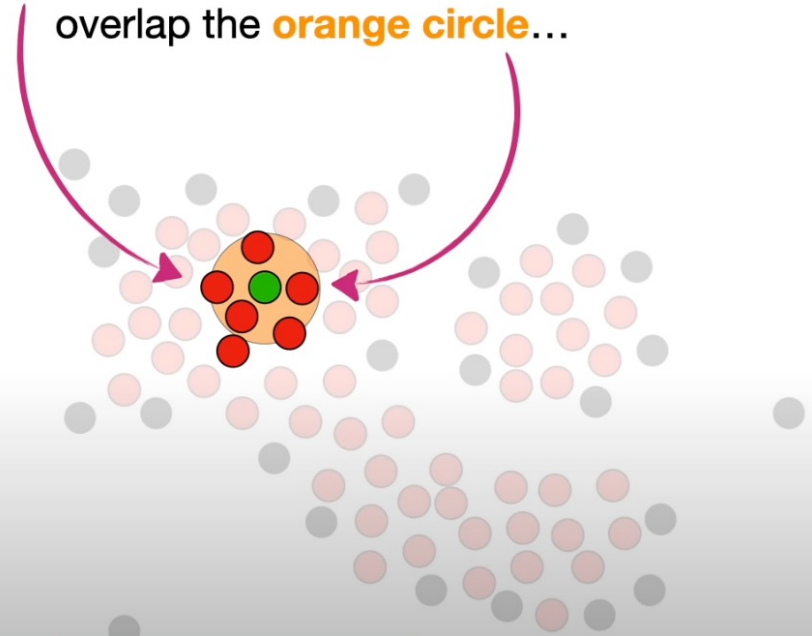
Ultimately, we can call all of these **red points** **Core Points** because they are all close to 4 or more other points...



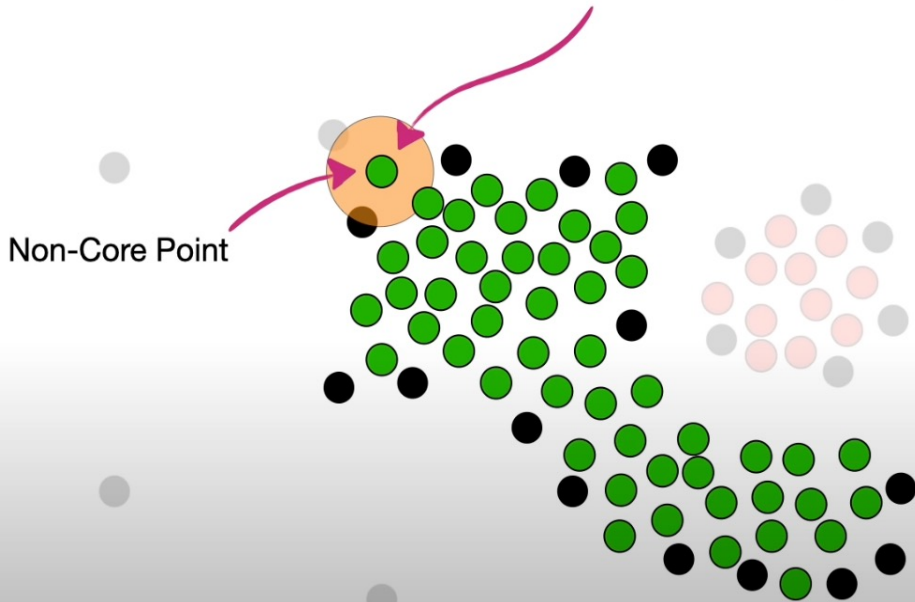
Now we randomly pick
a **Core Point**...



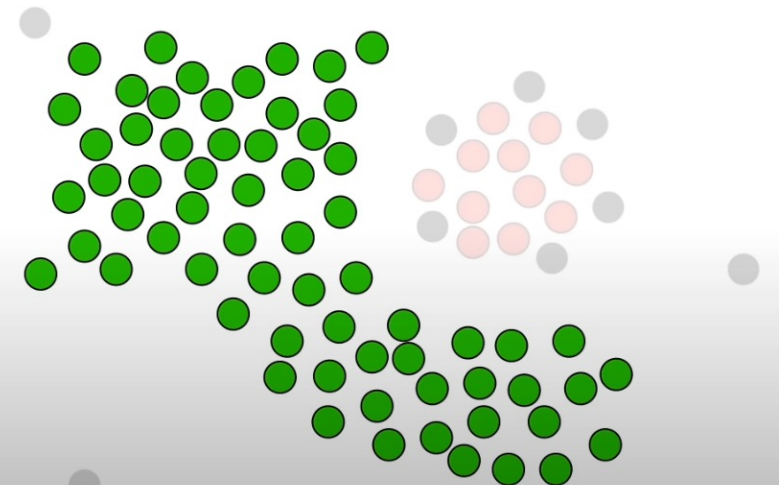
Next, the **Core Points** that are close
to the **first cluster**, meaning they
overlap the **orange circle**...



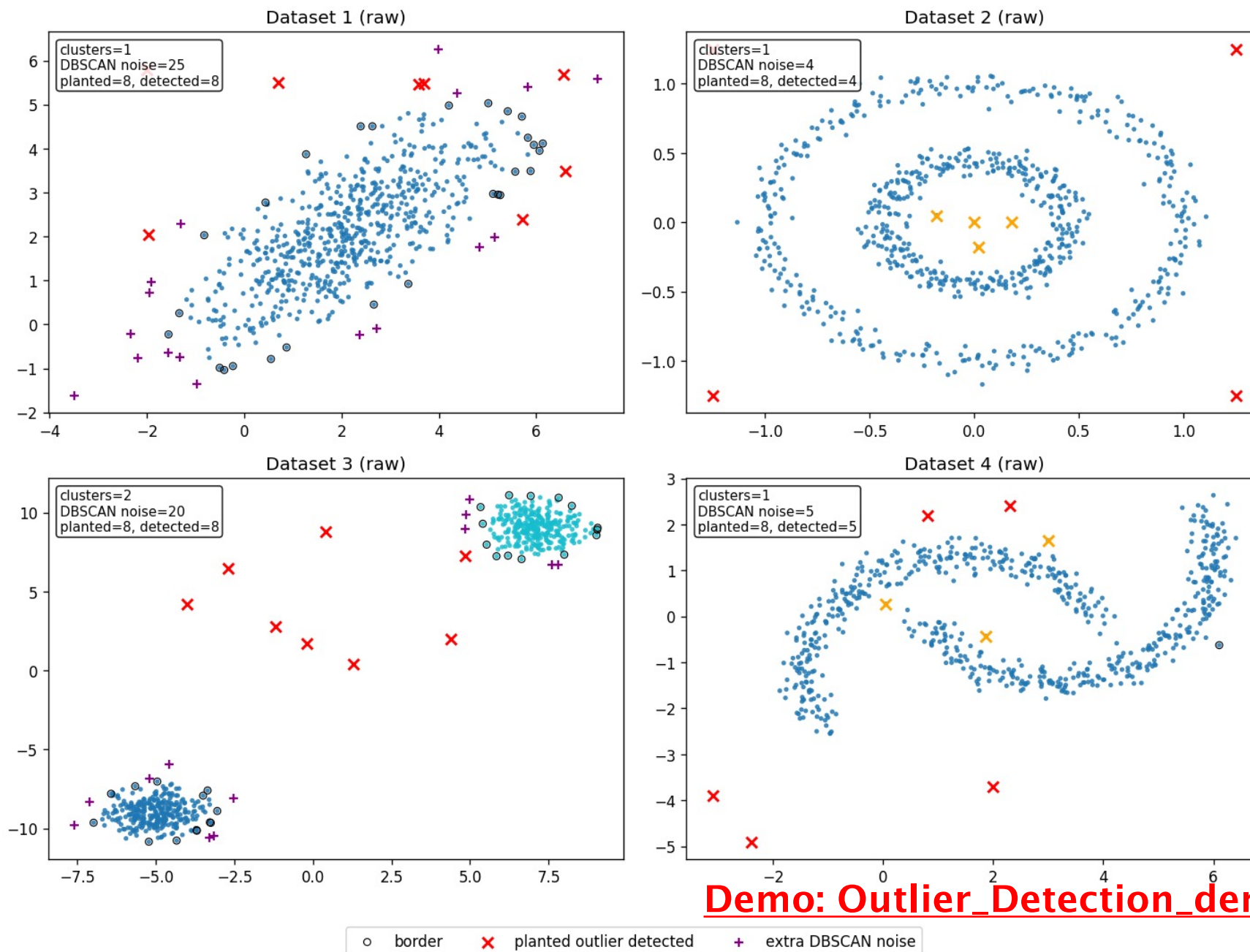
However, because this is not a **Core Point**, we do
not use it to extend the **first cluster** any further.



And now we are done
creating the **first cluster**.



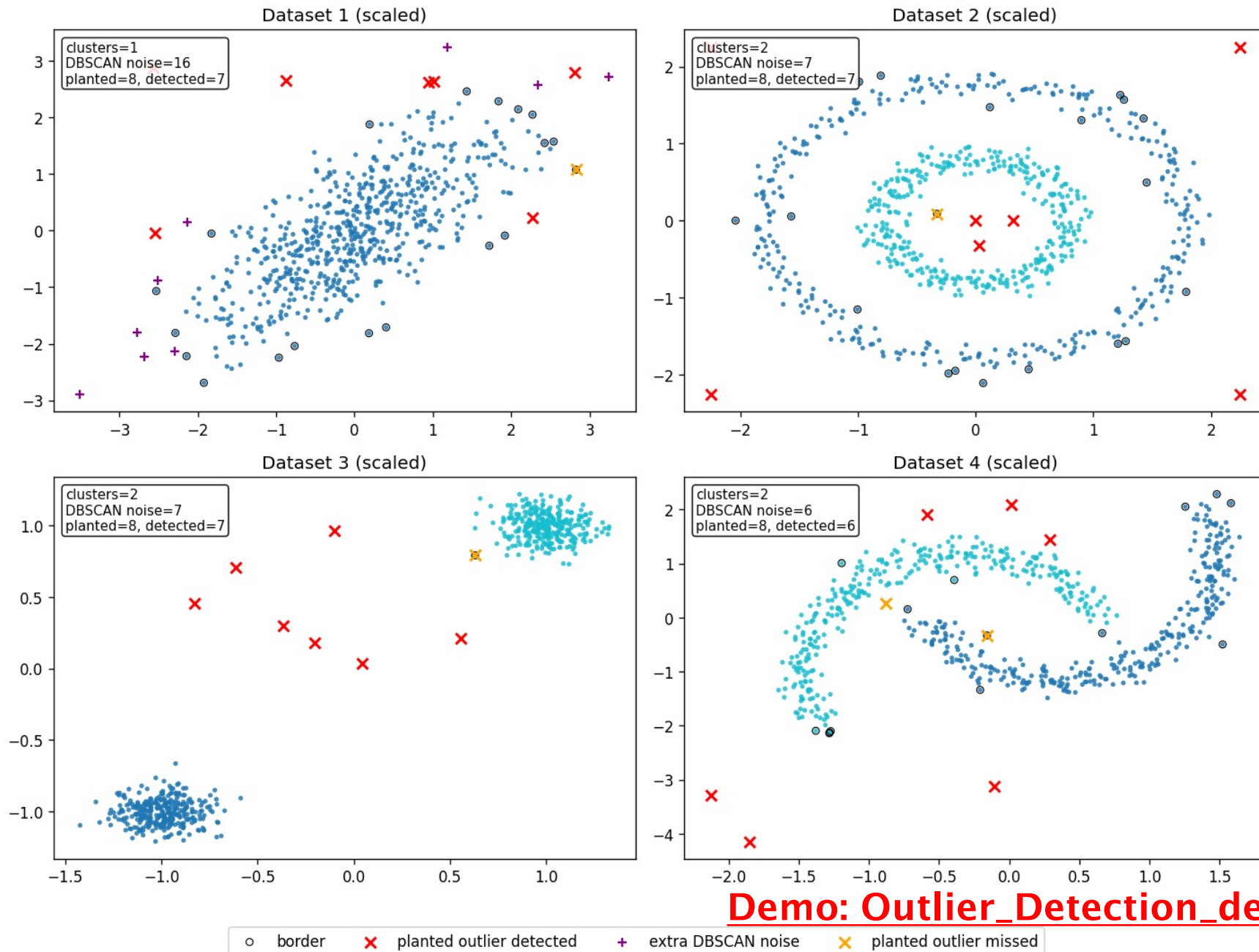
Outlier Detection – DBSCAN



Demo: Outlier_Detection_demo 18/20

Normalisation! - and adjusting *eps*

Outlier Detection – DBSCAN



Outlier Detection – Summary

Outlier detection is explored as a data mining problem:.

Extreme value statistics:

- ▶ to help tell the difference between an anomaly and an extreme member of a distribution

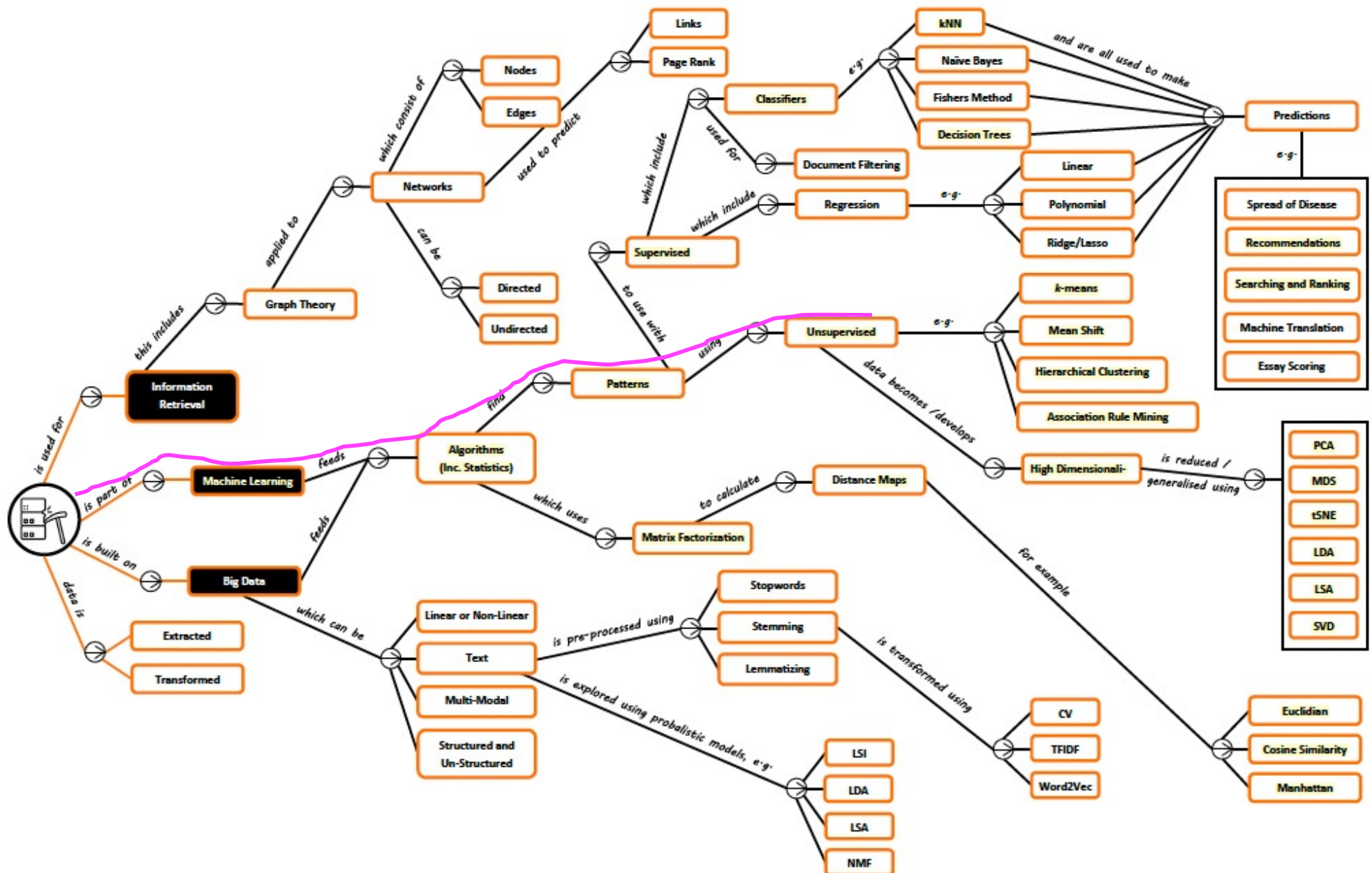
Gaussian Mixture Models:

- ▶ Models the system as a mixture of Gaussian distributions
- ▶ uses Expectation Maximisation to find parameters
- ▶ can be distorted by outliers

DBSCAN:

- ▶ Used for outlier detection
- ▶ Robust to outliers
- ▶ can have issues with parameters ϵ

Outlier Detection – Roadmap



Outlier Detection – Textbook

9 Anomaly Detection

*In anomaly detection, the goal is to find objects that do not conform to normal patterns or behavior. Often, anomalous objects are known as **outliers**, since, on a scatter plot of the data, they lie far away from other data points. Anomaly detection is also known as **deviation detection**, because anomalous objects have attribute values that deviate significantly from the expected or typical attribute values, or as **exception mining**, because anomalies are exceptional in some sense. In this chapter, we will mostly use the terms anomaly or outlier. There are a variety of anomaly detection approaches from several areas, including statistics, machine learning, and data mining. All try to capture the idea that an anomalous data object is unusual or in some way inconsistent with other objects.*

- ▶ Introduction to Data Mining, *P. Tan et al*
<https://www-users.cse.umn.edu/~kumar001/dmbook/index.php>

Outlier Detection – Learning Outcomes

- **LO1:** Demonstrate an understanding of techniques for outlier detection, such as: (exam)
 - ❖ Applying extreme value analysis method
 - ❖ Understanding the key idea and steps of the learned clustering methods for outlier detection
 - ❖ Discussing the advantages and disadvantages of the learned outlier detection approaches
- **LO2:** Implement the learned algorithms for outlier detection (coursework)

Assessment hints: Multi-choice Questions (single answer: concepts, calculation etc)

- *Textbook Exercises: textbooks (Programming + Mining)*
- *Other Exercises: <https://www-users.cse.umn.edu/~kumar001/dmbook/sol.pdf>*
- *ChatGPT or other AI-based techs*